

AI-Powered Civic Governance Platform

- 📄 Executive Summary
- 📄 Problem Statement
- 📄 Objectives
- 📄 Existing System & Limitations
- 📄 Proposed AI-Powered Civic Governance Platform
- 📄 Functional Requirements
- 📄 Non-Functional Requirements
- 📄 Complete System Workflow
- 📄 AI Architecture
- 📄 AI Model Selection
- 📄 Data Flow Diagrams (DFD Level 0–2)
- 📄 UML Diagrams
 - Use Case
 - Activity
 - Sequence
 - Class Diagram
- 📄 ER Diagram
- 📄 Database Schema (PostgreSQL + PostGIS)
- 📄 REST API Documentation
- 📄 Authentication & Security
- 📄 Privacy & Consent Framework
- 📄 Complaint Lifecycle
- 📄 Duplicate Detection Architecture
- 📄 Smart Routing Engine
- 📄 SLA Engine
- 📄 Escalation Engine
- 📄 Officer Workflow
- 📄 Citizen Workflow
- 📄 Analytics Dashboard
- 📄 GIS Integration

- ☐ Multilingual Support
- ☐ Voice/Image Processing Pipeline
- ☐ Explainable AI
- ☐ Accessibility Features
- ☐ Technology Stack
- ☐ Folder Structure
- ☐ Microservice Architecture
- ☐ Deployment using Docker
- ☐ CI/CD Pipeline
- ☐ Testing Strategy
- ☐ Future Scope
- ☐ Hackathon Innovation Points
- ☐ Presentation Script
- ☐ Complete Project Report (40–60 pages)

1. Executive Summary

1.1 Introduction

Rapid urbanization has significantly increased the number of civic issues reported by citizens, including potholes, overflowing garbage bins, water leakages, broken streetlights, drainage blockages, pollution, and public safety hazards. While many municipalities have adopted online grievance systems, most existing platforms function only as digital complaint registers and still rely heavily on manual intervention.

These traditional systems face several challenges:

- Multiple duplicate complaints for the same issue
- Incorrect department assignment
- Slow manual verification
- Lack of transparency in complaint resolution
- Limited multilingual support
- Poor accessibility for non-technical users
- No intelligent prioritization of emergencies
- Limited data-driven decision making

As a result, government departments receive thousands of fragmented complaints that require significant manual effort before action can begin. This delays issue resolution and reduces citizen trust.

The **Community Redressal Planner** is designed to overcome these limitations by introducing Artificial Intelligence into every stage of the civic grievance lifecycle. Rather than serving as a simple complaint registration portal, the platform acts as an **AI-powered Civic Governance System** that automates complaint analysis, intelligently routes incidents, monitors service-level agreements (SLAs), protects citizen privacy, and provides actionable insights to municipal authorities.

The platform supports multilingual complaint submission through text, voice, images, and messaging channels. AI models automatically classify issues, extract relevant entities such as locations and landmarks, detect duplicate complaints, assign priorities, recommend the appropriate government department, and continuously monitor progress until resolution.

By combining AI, GIS mapping, workflow automation, and explainable decision-making, the system enables faster response times, improves accountability, and enhances citizen satisfaction while reducing administrative workload.

1.2 Vision

To build an intelligent, transparent, multilingual, and citizen-centric civic governance platform that leverages Artificial Intelligence to improve public service delivery, optimize municipal operations, and strengthen trust between citizens and government agencies.

1.3 Mission

The mission of the Community Redressal Planner is to transform traditional grievance management into a proactive governance ecosystem by:

- Simplifying complaint submission across multiple channels.
 - Eliminating duplicate complaints through AI.
 - Automatically identifying the responsible government department.
 - Prioritizing complaints based on urgency and public safety.
 - Ensuring timely resolution through SLA monitoring and automated escalation.
 - Providing transparent complaint tracking for citizens.
 - Delivering analytics and decision support for government authorities.
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1.4 Project Objectives

The primary objectives of the platform are:

Intelligent Complaint Processing

Automatically classify complaints using Natural Language Processing (NLP) without requiring manual categorization.

Duplicate Detection

Identify semantically similar complaints submitted in different languages or formats and merge them into a single master incident while keeping all reporters informed.

Smart Department Routing

Assign complaints to the correct government department using AI confidence scores and rule-based validation.

Priority Prediction

Determine complaint severity based on contextual information and public safety risk.

Multilingual Accessibility

Support complaint submission in multiple Indian languages, including text and voice.

Real-Time Tracking

Provide complete transparency through complaint timelines, notifications, and status updates.

SLA Compliance

Monitor department-specific service deadlines and automatically escalate overdue complaints.

Data-Driven Governance

Generate dashboards, heatmaps, performance reports, and predictive analytics for municipal authorities.

1.5 Scope of the Project

The Community Redressal Planner supports the complete civic complaint lifecycle.

Citizen Services

- User registration and authentication
- Complaint submission via text
- Voice complaint submission
- Image upload
- GPS location capture
- Complaint tracking
- Appeal requests
- Notifications
- Complaint history
- Privacy controls

AI Services

- Language detection
- Translation
- Complaint classification
- Named Entity Recognition
- Duplicate detection
- Image analysis
- Speech-to-text conversion
- Priority prediction
- Department recommendation
- AI-generated complaint summaries

Government Services

- Officer dashboard
- Complaint assignment
- Workload balancing
- GIS visualization
- SLA monitoring

- Escalation management
- Analytics dashboard
- Audit logs

Administrative Services

- Department management
 - Officer management
 - Service directory
 - SLA configuration
 - AI confidence monitoring
 - System reports
 - User management
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1.6 Key Innovations

Unlike conventional complaint portals, the proposed platform introduces several AI-driven capabilities:

- AI-powered multilingual complaint understanding.
 - Semantic duplicate detection across languages.
 - Intelligent department recommendation.
 - Explainable AI with confidence scores.
 - GIS-based complaint visualization.
 - Automatic SLA tracking and escalation.
 - Real-time citizen updates.
 - Privacy-preserving data redaction.
 - AI-assisted officer recommendations.
 - Predictive hotspot analysis for recurring civic issues.
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1.7 Expected Outcomes

Implementation of the proposed platform is expected to:

- Reduce duplicate complaints by over **70%** through semantic matching.
- Decrease manual routing efforts by approximately **80%**.
- Improve average complaint resolution time through automated workflows.

- Increase transparency by providing real-time tracking and audit trails.
 - Enhance accessibility for citizens through multilingual and voice-based interfaces.
 - Enable municipal authorities to make evidence-based decisions using analytics and GIS insights.
 - Improve overall citizen satisfaction and trust in public service delivery.
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1.8 Stakeholders

The platform serves multiple stakeholders:

Stakeholder	Role
Citizens	Report issues, track progress, submit appeals
Municipal Officers	Resolve complaints and update status
Department Heads	Monitor workload, SLA compliance, and escalations
Municipal Administrators	Manage departments, users, and policies
Government Authorities	Analyze city-wide trends and performance
AI Services	Automate classification, routing, prioritization, and duplicate detection

1.9 Project Highlights

The proposed solution is designed as an **AI-powered Civic Operating System**, integrating complaint management, workflow automation, multilingual AI, geospatial intelligence, and governance analytics into a unified platform. It emphasizes transparency, accountability, fairness, accessibility, and measurable improvements in public service delivery while maintaining human oversight for critical decisions.

2. Problem Statement

2.1 Background

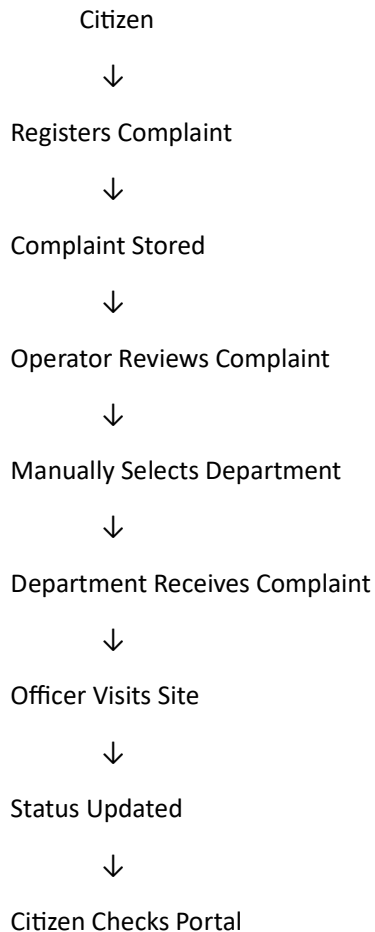
Modern cities rely on efficient public infrastructure and municipal services to ensure a high quality of life. Citizens frequently encounter issues such as potholes, overflowing garbage bins, broken streetlights, water leakages, blocked drainage systems, illegal construction, and public safety hazards. Timely reporting and resolution of these issues are essential for maintaining urban infrastructure and public trust.

Most municipal corporations provide online grievance portals where citizens can submit complaints. However, these platforms primarily serve as digital complaint registers. They collect complaints but lack the intelligence required to process, prioritize, and route them efficiently. As urban populations grow, municipalities receive thousands of complaints every day. Manual handling of these complaints results in delays, misclassification, duplicate records, and poor service delivery.

The need for an intelligent, automated, and transparent grievance management system has therefore become increasingly important.

2.2 Existing Complaint Resolution Process

The current workflow in most municipalities is shown below.



This process depends heavily on manual review and lacks intelligent automation.

2.3 Problems in Existing Systems

Although digital grievance portals have replaced paper-based systems, they still suffer from several operational and technical limitations.

2.3.1 Duplicate Complaints

The same civic issue is often reported by multiple citizens using different words, languages, or descriptions.

Example:

Citizen A-Street light not working.

Citizen B-Pole light is off.

Citizen C-स्ट्रीट लाईट बंद आहे.

Citizen D-Lamp near bus stand is damaged.

The existing system creates four separate complaints instead of recognizing that all refer to the same streetlight.

Consequences:

- Multiple officers receive duplicate work.
- Resources are wasted.
- Resolution statistics become inaccurate.
- Citizens receive inconsistent updates.

2.3.2 Incorrect Department Assignment

Most portals require citizens to select the appropriate department.

However, many citizens do not know which department is responsible.

Example:

A water leakage complaint may be submitted to:

- Road Department
- Sanitation Department
- Electricity Department

instead of the Water Supply Department.

Incorrect routing causes delays before the complaint reaches the correct authority.

2.3.3 Manual Complaint Classification

Government operators often read every complaint and manually categorize it.

Problems include:

- Time-consuming process
- Human errors
- Inconsistent categorization
- Delayed response

During emergencies or heavy complaint volumes, manual processing becomes a bottleneck.

2.3.4 Lack of Priority Assessment

Most systems process complaints on a first-come, first-served basis.

However, not all complaints have the same urgency.

Example:

Complaint	Actual Priority
Broken bench	Low
Garbage overflow	Medium
Water leakage	High
Live electric wire	Critical

Without intelligent prioritization, life-threatening incidents may not receive immediate attention.

2.3.5 Limited Transparency

Citizens often receive only basic status updates such as:

- Pending
- In Progress
- Resolved

They cannot see:

- Assigned officer
- Expected completion date
- Department handling the issue
- Resolution timeline
- Reasons for delays

This lack of transparency reduces public trust.

2.3.6 Poor Multilingual Support

India is linguistically diverse, yet many grievance portals support only English.

Citizens may submit complaints in:

- Marathi
- Hindi
- Tamil
- Telugu
- Kannada
- Bengali

Manual translation slows the process and may introduce errors.

2.3.7 Weak Accessibility

Existing systems often exclude:

- Senior citizens
- Visually impaired users
- Illiterate citizens
- People without smartphones
- Citizens in rural areas with poor internet connectivity

They typically lack:

- Voice input
 - Screen reader compatibility
 - Offline submission
 - Regional language interfaces
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2.3.8 No Intelligent Analytics

Authorities usually view only basic statistics such as:

- Total complaints
- Resolved complaints
- Pending complaints

They cannot easily identify:

- Complaint hotspots
- Frequently recurring issues
- Underperforming departments
- SLA violations
- Seasonal trends

As a result, decision-making remains reactive instead of proactive.

2.4 Root Cause Analysis

Problem	Root Cause
Duplicate complaints	No semantic similarity detection
Wrong department	Manual citizen selection
Delayed routing	Human verification
Long resolution times	Lack of automation
Poor transparency	Limited tracking mechanisms
Language barriers	English-centric interfaces
Accessibility issues	No inclusive design
Weak analytics	Lack of AI and GIS integration

2.5 Gap Analysis

Existing System	Proposed AI Platform
Manual complaint review	AI-based complaint understanding
Manual department selection	Intelligent department routing
Duplicate complaints	Semantic duplicate detection
Text-only complaints	Text, voice, image, WhatsApp, SMS
Single language	Multilingual support
Manual prioritization	AI priority prediction
Static dashboards	Real-time analytics and GIS heatmaps
Limited updates	Transparent tracking with notifications
Fixed workflow	Automated SLA monitoring and escalation

2.6 Why Artificial Intelligence is Needed

Artificial Intelligence enables the system to process complaints intelligently rather than simply storing them.

AI can:

- Understand complaints written in different languages.
- Detect the actual civic issue regardless of wording.
- Identify duplicate complaints using semantic similarity.
- Predict complaint urgency.
- Recommend the responsible department.
- Extract locations and landmarks automatically.
- Generate concise summaries for officers.
- Monitor SLA compliance.
- Support decision-making through predictive analytics.

This reduces manual effort while improving speed, consistency, and accuracy.

2.7 Real-World Scenario

Current System

A large pothole develops near a bus stand.

Within one day:

Citizen 1: Road broken near bus stand.

Citizen 2: Huge pothole outside the station.

Citizen 3: बस स्टँडसमोर खड्डा आहे.

Citizen 4: uploads an image.

The existing system creates **four independent complaints**.

Different officers may be assigned, leading to duplicated effort and fragmented updates.

Proposed AI System

The AI platform processes each submission by:

1. Detecting the language.
2. Translating the content (if required).

3. Classifying the issue as **Road Damage**.
4. Extracting the location ("Bus Stand").
5. Comparing the complaint with existing incidents.
6. Identifying all four reports as the same issue.
7. Creating **one master incident**.
8. Linking all four citizens to that incident.
9. Assigning it to the **Public Works Department (PWD)**.
10. Tracking SLA compliance and notifying all reporters of progress.

This approach reduces redundant work, accelerates resolution, and improves transparency.

2.8 Problem Definition

The challenge is to design an intelligent civic governance platform that can:

- Accept complaints from multiple channels (text, voice, images, messaging).
 - Support multiple Indian languages.
 - Automatically classify civic issues.
 - Detect duplicate complaints across different languages and wording.
 - Route complaints to the appropriate department.
 - Predict priority based on severity.
 - Monitor resolution through SLA enforcement.
 - Provide transparent tracking for citizens.
 - Deliver actionable analytics for municipal authorities.
 - Ensure privacy, security, and human oversight in AI-assisted decisions.
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2.9 Chapter Summary

The analysis of existing grievance systems reveals that current solutions are largely reactive and dependent on manual processes, resulting in inefficiencies, delayed responses, and limited transparency. The identified gaps highlight the need for an AI-driven platform capable of automating complaint processing, supporting multilingual communication, detecting duplicate issues, enabling smart routing, and providing data-driven governance.

3. Proposed System

3.1 Introduction

The **Community Redressal Planner** is an **AI-powered Civic Governance Platform** designed to modernize the way municipal complaints are received, processed, and resolved. Unlike traditional complaint portals that only register grievances, the proposed system automates the entire complaint lifecycle using Artificial Intelligence, Natural Language Processing (NLP), Computer Vision, Geographic Information Systems (GIS), and workflow automation.

The platform enables citizens to report civic issues through multiple channels, while AI intelligently analyzes the complaint, identifies duplicates, predicts urgency, assigns the correct department, and continuously monitors progress until resolution.

The system is designed around the principles of:

- Transparency
 - Accountability
 - Accessibility
 - Privacy
 - Explainable AI
 - Human Oversight
 - Efficient Public Service Delivery
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3.2 Vision of the Proposed System

To create an intelligent civic governance ecosystem where Artificial Intelligence assists municipal authorities in delivering faster, fairer, and more transparent public services while keeping citizens informed throughout the resolution process.

3.3 Objectives

The proposed platform aims to achieve the following objectives:

Primary Objectives

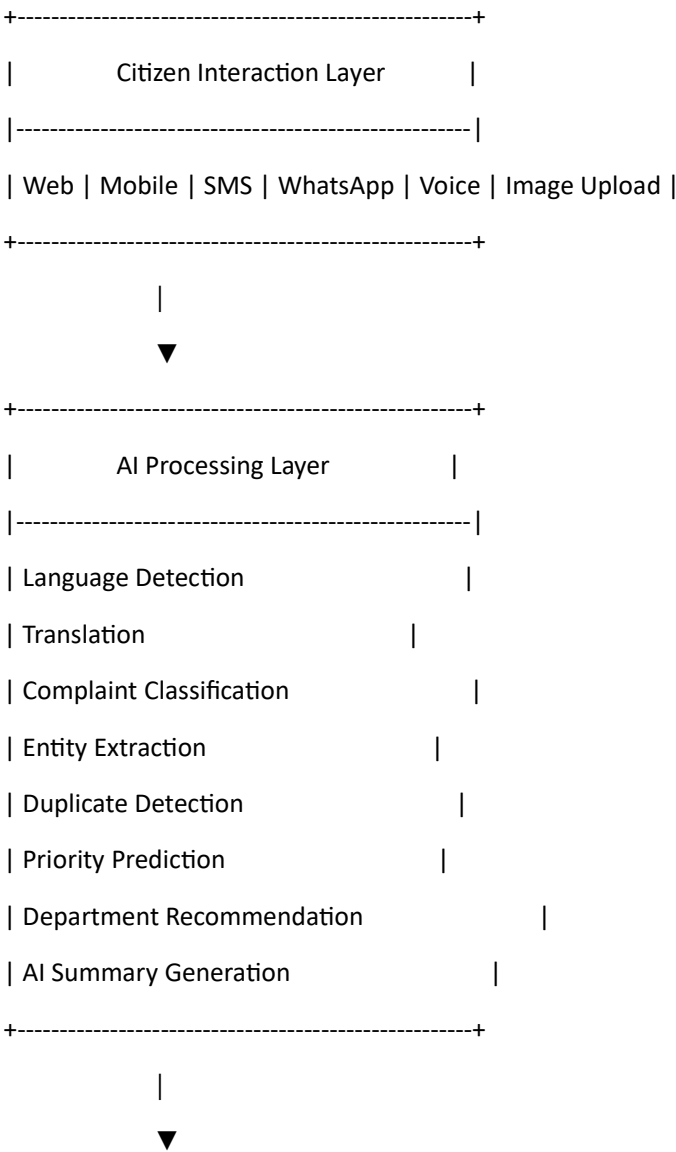
- Reduce complaint resolution time.
- Eliminate duplicate complaints.
- Improve department routing accuracy.
- Increase transparency for citizens.
- Enable multilingual complaint submission.
- Provide intelligent analytics for authorities.

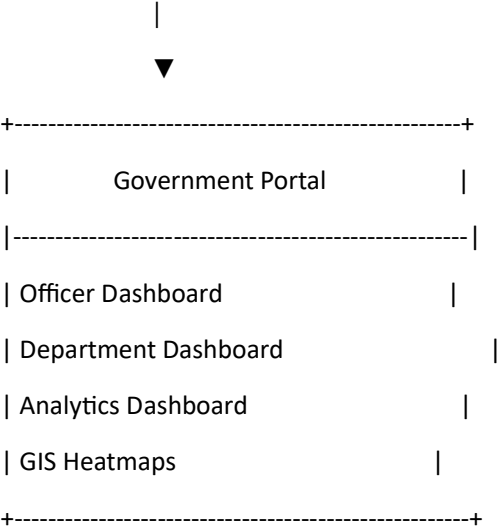
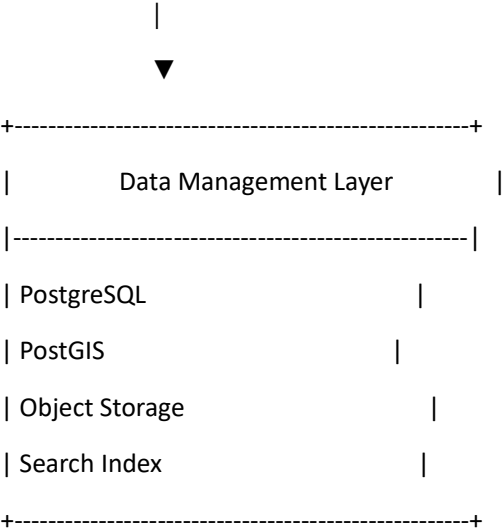
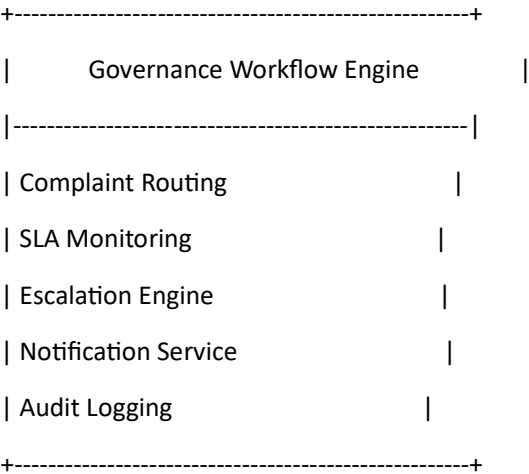
Secondary Objectives

- Reduce administrative workload.
- Increase citizen participation.
- Support digital governance initiatives.
- Improve resource allocation.
- Enable predictive maintenance.
- Enhance policy decision-making using analytics.

3.4 Proposed System Architecture

The proposed solution consists of six major layers.





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	Citizen Portal

	Complaint Tracking
	Timeline
	Notifications
	Appeals
	Privacy Settings
+-----+	

3.5 Major System Modules

The platform consists of the following modules.

1. Citizen Management

Responsible for:

- Registration
- Login
- Profile management
- Identity verification (OTP simulation)
- Complaint history
- Notification preferences

2. Complaint Intake Module

Accepts complaints through multiple channels:

- Website
- Mobile application
- SMS
- WhatsApp (simulation)
- Voice recording
- Audio upload
- Image upload

Each complaint receives a unique complaint ID.

3. AI Processing Module

This is the intelligence core of the platform.

It performs:

- Language Detection
- Translation
- Complaint Classification
- Entity Recognition
- Duplicate Detection
- Priority Prediction
- Department Recommendation
- AI-generated Summary

4. Complaint Workflow Engine

Responsible for:

- Complaint creation
- Incident management
- Routing
- Assignment
- Status updates
- Escalation
- Resolution
- Appeals

5. SLA Management

Every complaint category receives a predefined Service Level Agreement.

Example:

Category	SLA
Garbage	12 Hours
Streetlight	24 Hours
Water Leakage	48 Hours
Road Damage	7 Days
Illegal Construction	15 Days

If an SLA expires:

- Officer notified
 - Supervisor notified
 - Escalation created
 - Audit log updated
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6. Notification Module

Sends updates using:

- Email
- SMS
- Push Notification
- WhatsApp (simulation)

Citizens receive notifications for:

- Complaint accepted
 - Department assigned
 - Status changed
 - Resolution completed
 - Appeal decisions
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7. Analytics Module

Provides decision support through:

- Complaint trends
 - GIS heatmaps
 - Resolution performance
 - Department efficiency
 - SLA compliance
 - Citizen satisfaction
 - Recurring issue detection
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3.6 User Roles

The system supports multiple user roles.

Citizen

Functions:

- Submit complaint
 - Upload images
 - Record voice complaint
 - Track complaints
 - Receive notifications
 - Submit appeals
 - Manage privacy settings
-

Officer

Functions:

- View assigned complaints
 - Update progress
 - Upload work evidence
 - Resolve complaints
 - View duplicate suggestions
 - Monitor SLA
-

Department Head

Functions:

- Assign officers
 - Monitor workload
 - Approve escalations
 - View department reports
 - Track KPIs
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Municipal Administrator

Functions:

- Manage departments
 - Configure SLA
 - Manage officers
 - View city-wide analytics
 - Monitor AI performance
 - Generate reports
-

3.7 Complaint Lifecycle

Citizen



Submit Complaint



AI Receives Complaint



Language Detection



Translation (if needed)



Complaint Classification



Entity Extraction



Duplicate Detection



Priority Prediction



Department Recommendation



Human Review (if AI confidence is low)



Department Assignment



Officer Assignment



Field Verification



Resolution



Citizen Feedback



Case Closed

3.8 Functional Features

The proposed platform provides:

Complaint Management

- Complaint registration
 - Complaint editing
 - Complaint cancellation
 - Complaint tracking
 - Appeals
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AI Features

- Language detection
 - Complaint classification
 - Duplicate detection
 - AI summaries
 - Location extraction
 - Priority prediction
 - Department routing
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Government Features

- Officer dashboard
- SLA monitoring
- GIS dashboard

- Analytics
 - Reports
 - Escalation management
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Citizen Features

- Timeline tracking
 - Complaint history
 - Notifications
 - Privacy controls
 - Feedback system
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3.9 Advantages of the Proposed System

Existing System	Proposed System
Manual routing	AI routing
Duplicate complaints	Duplicate merging
Single language	Multilingual
No AI	Intelligent automation
Manual prioritization	AI priority prediction
Basic tracking	Real-time timeline
Static reports	Interactive analytics
Limited transparency	End-to-end visibility

3.10 Expected Benefits

Citizens

- Faster complaint resolution
- Easy multilingual reporting
- Real-time updates
- Transparent workflow
- Better trust in public services

Officers

- Reduced manual sorting
- Smarter workload distribution
- AI-assisted decision support

- Better SLA compliance

Government

- Reduced operational costs
 - Improved resource allocation
 - Actionable analytics
 - Better policy planning
 - Higher citizen satisfaction
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3.11 Assumptions

The system assumes:

- Citizens provide accurate complaint details.
 - Departments regularly update complaint status.
 - Internet connectivity is available for online services.
 - AI models are periodically retrained.
 - Human review is available for low-confidence AI decisions.
 - GIS location data is reasonably accurate.
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3.12 Constraints

- AI predictions may occasionally require human verification.
 - Voice quality affects speech recognition accuracy.
 - Image quality influences object detection performance.
 - Regional language dialects may reduce translation accuracy.
 - Integration with real government systems is outside the scope of the hackathon prototype.
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3.13 Chapter Summary

The proposed **Community Redressal Planner** transforms a traditional grievance portal into a complete **AI-powered Civic Operating System**. By combining multilingual complaint intake, AI-driven analysis, automated workflows, SLA monitoring, and analytics, the platform improves efficiency, transparency, and accountability while maintaining human oversight for critical decisions.